

University Management System

1. Integrated Human Resource Management System
2. Academics Management System
3. Electronic Filing Management System
4. Standalone API based backend for role-based access control, secure API access for 3rd Party apps and other university applications

Specifications:

SI No.	Item	Description	Remarks
1	Integrated Module - Authentication & User Management	A foundational secure login, multi-factor support, profile management, and dynamic role switching	The login system should be shared across all systems and modules developed under this programme and must be scalable to support all future University modules and systems
1.1	Integrated Module - Login/Signout/ Password Reset & Account Recovery Module	Email/Phone number/Admin Approval based access systems	Modern Login/Signout/ Password Reset & Account Recovery system supporting OAuth, ReCaptcha and Error Validation and Failed Login handling enabled with detection mechanism for malicious intent. API based implementation for ease of integration of 3 rd Party apps and other university applications
2	Integrated Module - Human Resource Management System	An independent database system for all stakeholders of the University	KYC validated management system with API exposed to enable integration of 3 rd Party apps and systems to support necessary verification/validation/authentication of all stakeholders
2.1	Integrated Module Role Management System	Dynamically switching roles for each user based on requirement and access	Permissions of each user roles should be accessible with ease. Different dashboards for each type of user role are to be implemented for transition between roles. User Sessions have to be securely maintained and separate access control is to be provided for changing of Roles
3	Integrated Module Session & Security	Enterprise-grade security mechanisms, including encrypted communication, role-based	The application/platform shall implement industry-standard security protocols including JWT (JSON Web Token) Bearer Token–based authentication and authorization mechanisms to ensure secure access control,

		access control, secure API access, session security, and protection against unauthorized access and cyber threats.	session management, and API communication. The system shall support role-based access control (RBAC), encrypted data transmission over HTTPS/SSL, secure credential handling, token expiration and refresh policies, and protection against unauthorized access, tampering, replay attacks, and common web vulnerabilities in compliance with prevailing cybersecurity best practices.
4	AMS - Academic Calendar Management	A comprehensive system defining, managing, and tracking academic sessions, semesters, examination schedules, holidays, registration timelines, and other institutional academic activities.	All course offerings, enrollments, grade sheets, and results shall be mapped and managed based on the respective Academic Calendar and Semester combination. Academic Cell reserves the rights to create/ modify academic calendar with notifications and updates to be sent to all stakeholders.
5	AMS - Course Management module	Creation, modification, and management of course details including Course Number, Course Title, Department, Credit Structure, Course Type, Program Level, Course Status and other necessary attributes.	A centralized course data management with validation controls, role-based access for students and faculties, audit tracking, and integration with academic, examination, and enrollment workflows.
5.1	AMS - Course Offering module	Creation, publication, and management of semester-wise course offerings	Workflow-based course offering processes with dropdown-based master data mapping, auto-population of course details, multi-faculty assignment, enrollment

		with fields including Academic Calendar, Semester, Course Selection, Maximum Enrollment Limit, Teacher Assignment, Section/Batch, Practical Group, and Offering Status	control, status-based publishing, and seamless integration with student enrollment and academic management systems.
5.2	AMS - Credit Structure system	All credit formats available should be supported	All combinations has to be supported following University norms of Theory/Practical breakdown such as: 2+0, 0+2, 1+1, 0+1, 3+0.
5.3	AMS - Student Enrollment Module	The system should be able to handle student data and information from application to university to selection and finally enrolment	This module should seamlessly able to gather information from the university application system and bridge information from Student Human Resource Management system and Academic Cell for enrolment of the student into courses
5.4	AMS - Teacher Enrollment Management	This system should enable ease of reviewing, approving, and rejecting student course enrollment requests on a course-wise basis.	The module shall support bulk and individual approval workflows, mandatory rejection remarks, automated status notifications via system and email, and automatic synchronization of approved enrollments with course-wise grade sheet generation.
5.5	AMS - Tabulation Module and automated GPA calculator	Grading and GPA Management module enabling faculty to generate course-wise grade sheets, record student assessments,	A fully standalone system for Tallying for data with special attention to ease of Tabulation. Automated grade computation, GPA/CGPA calculation logic, controlled mark entry validation, audit-enabled modification tracking, and multi-level approval workflows with role-based authorization prior to official result publication.

		and process result compilation based on a configurable 10-point grading framework.	
6	AMS - Grade Card Generation system	Through Grade Sheet Filters & Selection Process authorised user should be able to generate, verify and issue Grade Cards	the correct student list (only enrolled + approved students) and the correct marking has to be generated taking into consideration the Academic Calendar, Semester, Course Type and Grade Sheet Type
7	AMS - Multi-Stage Result Approval Workflow Module	This module should enable sequential validation, approval, and controlled publication of grade sheets and academic results through designated institutional authorities.	A configurable 5-stage approval pipeline with role-based access control, digital signature/OTP-based authentication, timestamped audit trails, stage-wise locking mechanisms, automated notifications, rejection rollback handling, and controlled result publication with downloadable PDF generation for students.
8	AMS - Digital Signature & Timestamp	a secure Digital Approval and Signature Verification mechanism for authenticated approval of grade sheets, academic records, and result processing workflows.	The module shall support 4-digit confirmation PIN and OTP-based signature authentication, immutable audit logging, timestamp and IP-based activity tracking, role-wise approval records, and automated embedding of digitally signed approval sections within generated PDF grade sheets and result documents.
9	AMS - Notifications and rejection handling	An automated Notification and Rejection Management module for workflow-driven communication, status	The module shall support real-time in-application and email notifications, stage-wise workflow escalation, mandatory rejection remarks, rollback-based resubmission handling, editable return states, and comprehensive

		escalation, and controlled handling of approvals and rejections.	audit trail maintenance capturing rejection history, approver details, timestamps, and action logs.
10	AMS - Postgraduate and Research Management System	A comprehensive module for administration of PG/PhD academic activities, advisory committee formation, research supervision, and student research lifecycle tracking.	The module shall support role-based advisory committee management including Major Advisor, Co-Major Advisor, and Committee Members, with faculty eligibility validation, workflow-driven committee formation, assignment acceptance, automated notifications, controlled committee locking, authorized modification handling, and secure access to student academic and research records.
11	AMS - Result Publication and Digital Document Management module	The system shall provide a secure module for controlled result dissemination, student result access, automated grade sheet generation, and issuance of digitally signed academic documents.	The module shall support role-based result publication workflows, SGPA/CGPA computation display, branded PDF generation with institutional letterhead and watermark, QR-based verification, unique document reference numbering, digital signatures, password-protected read-only documents, controlled student access, audit-enabled document storage, and automated email/in-app result notifications upon official release.
12	AMS - Audit Logging and Security Management module	A comprehensive monitoring, traceability, and immutable recording of all user activities, workflow actions, and administrative operations across the AMS platform.	The module shall support tamper-proof audit trails with event-level logging, IP and User-Agent capture, role-context tracking, entity-level activity mapping, JSON-based metadata storage, timestamp synchronization in IST, immutable read-only log architecture, configurable retention policies, and secure archival for compliance, accountability, and academic governance requirements.
13	eFMS - Command	A fully role-aware and	A docket based primary workspace. Lists all Active files currently

	Centre for Electronic File Management	dynamically renders widgets, sections, and navigation links based on the user's active role. It provides instant access to active files, search, tutorials, and role switching.	assigned to or pending action by this user. Should show file number, subject, from whom, received date, urgency badge, and days elapsed
14	eFMS – Docket Detail View System	The system shall provide a centralized Docket Entry and File Repository Management module for registration, tracking, classification, retrieval, and workflow-based movement of official files and correspondence.	The module shall support system-generated unique file numbering, metadata-driven file indexing, role-based file actions, SLA-based elapsed time indicators, category/type tagging, advanced multi-parameter filtering, keyword-based search, status tracking, and secure repository management with department-wise access control and audit-enabled file movement history.
15	eFMS – File Creation and File Number generation Module	A system module for initiating, categorizing, and uniquely registering official electronic files with workflow-ready notesheet generation.	The module shall support structured file creation forms, auto-generated immutable file numbering logic, department-wise sequential indexing, configurable category/type classification, metadata-driven validation, role-based visibility controls for confidential files, priority tagging for urgent files, searchable repository integration, and automated creation of linked digital notesheet workspaces.
16	eFMS – Trace My Files Module	Advanced search interface to allow users to search across all files they are authorized to see.	The module shall support advanced multi-parameter search with exact and partial matching, role-based visibility controls, metadata-indexed retrieval, date-range filtering, real-time file location tracking, summarized result views, and interactive visual file timeline

			integration for end-to-end workflow traceability.
17	eFMS – Notesheet Workspace	A centralized module for creation, review, editing, and collaborative management of official digital notesheets linked to workflow files within the eFMS platform.	The module shall support rich-text editing, version-controlled collaboration, role-based access permissions, real-time save functionality, structured notesheet history, audit-enabled activity tracking, timestamped modifications, and immutable logging of all user actions performed on the notesheet workspace.
17.1	eFMS – Notesheet Editor	The system shall provide an enterprise-grade Rich Text Editor integrated within the Notesheet Workspace to enable browser-based drafting, formatting, review, and preparation of official digital correspondence and notesheets.	The module shall support CKEditor 5/TinyMCE-based WYSIWYG editing, advanced text and paragraph formatting, MS Word-compatible copy-paste preservation, dynamic table management, server-side auto-save and draft restoration, full-screen editing mode, live word/character count, spell checking, find-and-replace functionality, multi-level undo/redo history, and print-preview rendering for PDF-ready document generation.
17.2	eFMS – Draft Auto-Save and Restore module	The system shall provide a secure system within the Notesheet Workspace to prevent data loss and ensure continuity of document preparation activities.	The module shall support silent server-side periodic auto-save operations, timestamp-based draft versioning, user-file linked draft persistence, interrupted session recovery detection, contextual restore/discard prompts, and secure restoration of unsaved editor states with audit-enabled draft lifecycle management.
17.3	eFMS – Attachment Management module	The module should provide an easy system for uploading, organizing, previewing, and managing supporting documents	The module shall support controlled file upload validation, configurable attachment size limits, multi-format document support, metadata-based attachment tagging, in-browser preview for supported file types, cumulative storage threshold enforcement, role-based attachment deletion

		associated with notesheets and workflow files.	permissions, and post-dispatch attachment locking with audit-enabled tracking of all attachment operations.
17.4	eFMS – Notesheet Version Control	The system shall provide a module for maintaining immutable historical snapshots of all notesheet activities throughout the document lifecycle.	The module shall support automatic version snapshot generation on save and dispatch events, timestamped version tracking, author and action attribution, read-only historical access, immutable audit-compliant storage, and side-by-side differential comparison with highlighted content changes between selected versions.
17.5	eFMS – Notesheet Viewing and Remarks Management module	The module shall provide a controlled system for handling received files and maintaining structured inter-departmental communication trails.	The module shall support read-only viewing of previously submitted notesheets, append-only remark entry, role-based forwarding and return actions, immutable preservation of prior remarks, user-wise attribution with designation and timestamp metadata, and sequential notesheet thread generation for complete workflow traceability and audit compliance.
17.6	eFMS - Recipient Selection Module	The system shall provide a module for secure routing, hierarchical recipient discovery, and controlled movement of files across departments and establishments within the eFMS platform.	The module shall support multi-level hierarchical recipient search, autocomplete-enabled filtering, employee code and name-based lookup, out-of-office status indicators, dispatch-based notesheet locking, workflow-driven docket transfer, and configurable favourite recipient management for expedited file routing and communication efficiency.
17.7	eFMS - File Dispatch and Return Management	The system shall provide a secure module for controlled routing, transfer, and lifecycle handling of notesheets and	The module shall support dispatch-triggered notesheet locking, hierarchical and favourite-based recipient selection, optional dispatch remarks and priority tagging, automated docket movement, notification generation, immutable dispatch logging, return-

		workflow files between authorized users.	based workflow reversal with mandatory remarks, controlled notesheet reactivation for editing, and complete audit tracking of all dispatch and return events.
17.8	eFMS - File Locking and Concurrent Access Control module	The system shall provide a secure module to ensure transactional integrity, controlled editing rights, and secure lifecycle management of workflow files and notesheets.	The module shall support state-based file locking mechanisms, single-user write access enforcement, automatic transition between draft, in-transit, returned, and archived states, inactivity-based lock timeout release, administrator-controlled lock override, and contextual read-only access notifications for concurrent access prevention and workflow consistency.
17.9	eFMS - File Tracking and Transparency Module	The system shall provide a module for real-time monitoring, traceability, and visualization of file movement across the eFMS workflow ecosystem.	The module shall support end-to-end file movement tracking, user and department-level activity mapping, timestamped action history, real-time workflow status visibility, visual timeline representation, and immutable audit-based trace records capturing all handling, forwarding, return, and processing events associated with each file.
17.10	eFMS - Visual File Timeline and Summary Management System	The system shall provide a module for chronological visualization, monitoring, and analytical overview of file lifecycle activities within the eFMS platform.	The module shall support vertically sequenced workflow timelines with color-coded status indicators, actor-wise action history, expandable notesheet and attachment views, timestamped workflow events, elapsed time analytics, return-state highlighting, and compact file summary dashboards displaying file metadata, current holder details, movement statistics, elapsed duration, and real-time workflow status.
17.11	eFMS - Advanced Search and Filtering Module	The system shall provide an module for efficient retrieval, categorization, and analysis of	The module shall support multi-parameter filtering based on financial year, file category, file type, workflow status, department, date range, creator identity, and keyword-based partial matching, with indexed search optimization,

		files and workflow records across the eFMS platform.	role-based data visibility, and real-time retrieval of authorized records.
17.12	eFMS - Notification Management module	The system shall provide a centralized module for real-time delivery of workflow, security, and system-generated alerts across in-application, email, and SMS communication channels.	The module shall support event-driven and role-aware notification workflows, configurable multi-channel alert delivery, SLA and inactivity-based triggers, priority escalation for urgent files, OTP and security notifications, system-wide broadcast announcements, and delivery logging with user-level notification status tracking.
17.13	eFMS - Audit Logging and Security Management	The system shall provide a comprehensive module for immutable recording, monitoring, and traceability of all user activities, workflow operations, and administrative actions across the eFMS platform.	The module shall support event-level audit capture with role-context awareness, IP and User-Agent tracking, entity-based activity mapping, JSON metadata storage for transactional details, timestamp synchronization in IST, immutable read-only log architecture, secure long-term archival with minimum 10-year retention, and tamper-proof compliance mechanisms ensuring legal accountability, operational transparency, and forensic traceability.
18	Integrated Module – API enabled development	The system shall provide an API system enabling secure interoperability, real-time data exchange, and seamless integration with external enterprise, academic, financial, authentication, and government systems through	The module shall support RESTful/API-first architecture, JWT/OAuth2-based authentication, secure API gateway integration, webhook/event-driven communication, third-party apps and systems of ERP/HRMS/SSO integration, rate limiting, API audit logging, encrypted payload transmission, and scalable microservice-compatible interoperability frameworks.

		standardized APIs.	
19	Integrated Module – AI Powered development	The system shall provide AI systems for intelligent automation, workflow optimization, document analysis, predictive insights, and smart assistance across the platform ecosystem.	The module shall support AI/ML-driven recommendations, intelligent document classification, semantic search, OCR-assisted data extraction, anomaly detection, workflow prioritization, natural language processing, AI-assisted drafting and summarization, predictive analytics dashboards, and configurable AI service integration with secure processing and audit compliance.
20	Integrated Module – Customization	The system shall provide flexibility for implementation of client-specific enhancements, workflow modifications, feature extensions, and configurable business logic adaptations across the platform.	The module shall support structured change request management, configurable customization workflows, impact and complexity assessment, version-controlled deployment, and commercial engagement models based on hourly effort or per-change-request costing depending on scope, technical complexity, and implementation requirements.

- Preference shall be given to bidders that demonstrate an active technical collaboration, formal research partnership, memorandum of understanding (MoU), sponsored research agreement, or other documented joint work relationship with recognized Educational Institutions, Universities, Indian Institutes of Technology (IITs), National Institutes of Technology (NITs), Government-funded research organisations, or Government-funded Innovation Centres.
- For the purpose of evaluation, "active collaboration" means (a) a signed agreement or MoU within the last three years; and/or (b) at least one joint technical activity within the last 24 months (examples: joint research project, joint patent filing, sponsored laboratory work, or faculty industry consultancy).
- Bidders must submit documentary evidence of the collaboration (signed MoU/ agreement, copy of joint publication or patent records, joint project reports, internship/placement records, or official letters from the partner institution). Documentary proof shall be submitted along with the Technical Bid and will be subject to verification.

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